Web-to-Warehouse Product Intelligence Pipeline

A compliance-first, orchestrated and testable ETL platform that turns permitted public web sources into a dimensional warehouse with full price history, duplicate resolution, catalog reconciliation and a measured data-quality posture.

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The problem

Manual collection is slow, inconsistent, and destroys the history that makes pricing decisions possible.

Slow

- Analyst hours spent copying prices into spreadsheets
- A refresh cycle measured in days, not minutes

Inconsistent

- Same product named three ways
- Currencies and formats never normalised

Amnesiac

- No history, so no trend
- No audit trail, so no defensible claim

The question nobody can answer: what did this product cost last March, and on whose authority?

Objective and scope

Collect permitted product data, standardise it, resolve duplicates, and load it into an analytical database that answers business questions.

In scope

- Scrape permitted structured product information
- Respect robots.txt, rate limits and website terms
- Clean names and categories; normalise currencies
- Detect duplicates and compare against the internal catalogue
- Load into PostgreSQL / MySQL with historical snapshots
- Identify price, new, removed and category changes with SQL
- Orchestrate with Airflow; expose the result over REST

Deliberately out of scope

- Bypassing any site's protections
- Personal or behavioural data
- Real-time streaming ingestion
- Training a machine-learning model
- Anything not evidenced by a test

The pipeline, end to end

Nine stages, orchestrated by Airflow, each reporting progress and each writing evidence of what it did.

extract → **stage** → **transform** → **resolve** → **load** → **detect** → **aggregate** → **reconcile** → **quality**

34	29	23	12
permitted sources	physical tables	analytical views	DQ rules

Every stage publishes progress to a leased job queue, so a slow run is visible rather than silent.

Compliance is evidence, not a claim

The ingestion layer is where a scraper earns the right to run.

Enforced in the fetch layer

- robots.txt parsed and its verdict recorded per request
- Per-source rate limits and crawl-delay honoured
- Conditional requests and an on-disk cache
- An identifying User-Agent with a contact address

Recorded as proof

- Every request logged with URL, status and elapsed time
- The robots verdict stored alongside each request
- Cache hits distinguished from live fetches
- One audit row per pipeline run, queryable by screen

A reviewer can reconstruct **what was fetched, when, and whether it was permitted** without reading the code.

Warehouse and schema ownership

A Kimball star schema, owned by a migration rather than by convention.

Dimensions

- dim_product
- dim_category
- dim_source
- dim_date
- dim_currency

Facts

- fact_price_snapshot — the historical price table
- fact_catalog_snapshot — our own list prices
- chg_price_change — detected movements
- chg_product_event — lifecycle

Aggregates

- agg_category_daily
- agg_brand_monthly
- Materialised during the run

Alembic creates all 29 tables and applies all 23 views; `alembic check` fails the build when the models and the migration disagree.

Beyond history: what the data says next

History is already in the fact table. These three things are not.

Forecasting

- Damped Holt-Winters with two-sided bands
- Accuracy measured on a held-out tail, never on the fit

Anomaly detection

- MAD, 3σ and IQR fences run together
- Two of three must agree, so an outlier cannot pass its own test

Pricing advice

- Category elasticity
- Raise, cut, hold or negotiate — with the reason stated

Where the history is too short, the API returns a flat series with wide bands rather than a confident-looking curve.

What the user actually gets

Operate

- Pipeline and run comparison
- Live job queue with cancel and retry
- Source health

Explore

- Query lab
- Aggregate builder
- Saved views

Decide

- Forecasts and anomalies
- Catalog reconciliation
- Reports as PDF or CSV

Govern

- Roles and scoped API keys
- Two-factor authentication
- A full audit trail

23
screens

222
API operations

666
catalogued features

5
themes, 12 accents

SSE
realtime updates

Verification

What has to pass before this counts as working.

Gate	Result
pytest (unit + integration)	619 passed
API smoke, against the running stack	104 passed
Ruff (app, tests, scripts, DAGs, migrations)	clean
mypy	clean, 126 files
TypeScript, ESLint at zero warnings, production build	clean
Render smoke test, every screen	23 screens
alembic check	no drift
Mermaid diagrams rendered	88 of 88
Documentation links resolved	1335

Structural figures are measured by `scripts/project_stats.py` and pinned by `scripts/check_stats.py`, which fails when a documented number stops matching the code.

Documentation as a deliverable

Planning

- Proposal
- Project plan
- Roles and RACI
- Risk register
- KPIs

Design

- Requirements
- System analysis
- Database design
- DFDs and behaviour diagrams
- UI/UX

Build

- API reference
- Testing strategy
- User manual
- Technical documentation
- Feature inventory

Subsystems

- Forecasting
- Jobs and realtime
- Reporting
- Migrations
- Security and access control

34
documents

88
Mermaid diagrams

36
static site pages

What made it hard, and what was done about it

Duplicate resolution

- The same product appears under three names on three sites
- Blocking keys to bound the comparison set
- A similarity threshold, with the strategy recorded per match

Rebuilding a schema in place

- create_all cannot express a change, a rollback or a history
- Alembic adopted; Airflow metadata moved to its own database

Realtime across workers

- An in-memory broker only sees its own process
- Events now also polled from a persisted log

The one that was **not** caught by any static check: an external store returning a new snapshot object per call, which put every screen into an infinite render loop. It is now pinned by a test.

Delivered

A working pipeline, a dimensional warehouse with history, a measured quality posture, a REST API, a dashboard, and documentation that a reviewer can verify rather than take on trust.

- Ingestion module
- Database schema
- Historical price table
- ETL pipeline
- Airflow DAG
- Data-quality checks
- SQL analysis
- Visualisation
- REST API
- Documentation